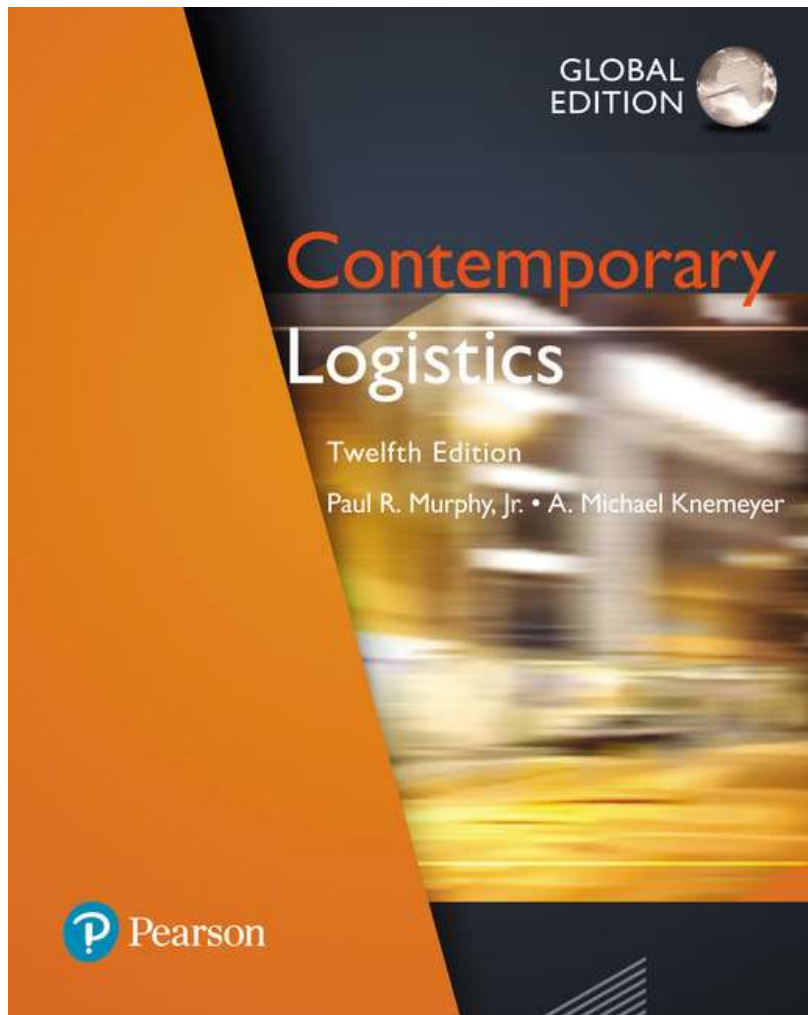


Contemporary Logistics

Twelfth Edition, Global Edition



Chapter 14

International Logistics

Learning Objectives

- 14.1** To discuss macroenvironmental influences on international logistics
- 14.2** To explain documentation as well as terms of sale and methods of payment for international shipments
- 14.3** To distinguish among the unique activities of international trade specialists
- 14.4** To illustrate transportation and inventory considerations in international distribution
- 14.5** To introduce you to the Logistics Performance Index

International Logistics (1 of 2)

- International logistics
 - Refers to logistics activities associated with goods that are sold across national boundaries

International Logistics (2 of 2)

- International logistics occurs when:
 - A firm exports a portion of a product made or grown
 - A firm imports raw materials
 - Goods are partially assembled in one country and then shipped to another, where they are further assembled or processed
 - A firm is global in outlook and sees almost all nations as markets, sources of supply, or sites for markets or for assembly operations
 - Because of geography, a nation's domestic commerce crosses foreign borders, often in bond. Goods going in transit through a country before ultimate destination

Macroenvironmental Influences on International Logistics (1 of 5)

- Macroenvironmental influences
 - Refer to the uncontrollable forces and conditions facing an organization and include **cultural**, demographic, **economic**, natural, **political**, and technological factors¹
 - For example, many countries have regulations and laws pertaining to air pollution; however, enforcement of them is not uniform from country to country

¹http://www.marketingpower.com/_layouts/Dictionary.aspx?dLetter=M.

Macroenvironmental Influences on International Logistics (2 of 5)

- Political factors
 - Political restrictions on international trade can take a variety of forms
 - Tariffs and taxes on import of certain items
 - Nontariff barriers
 - Import quota: Limits the amount or product (either in units or by value) that may be imported from any one country during a period of time. For example due to health and safety
 - Embargoes and Sanctions
 - The prohibition of trade between particular countries

Macroenvironmental Influences on International Logistics (3 of 5)

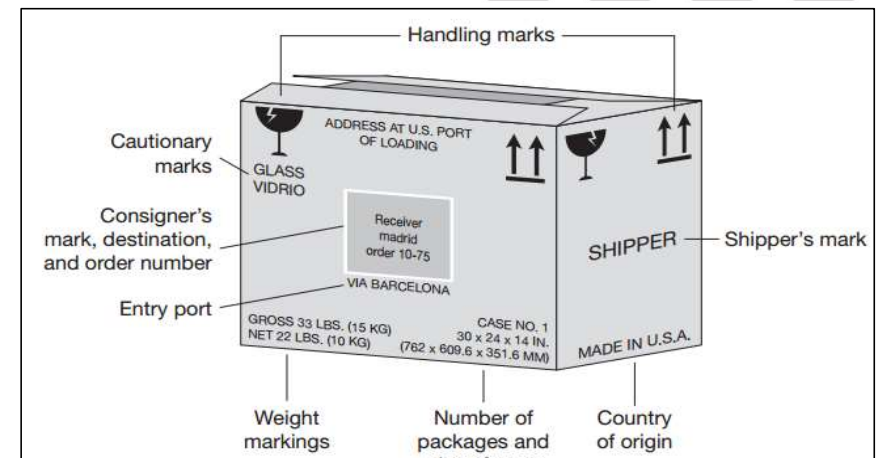
- Political factors
 - Degree of federal (central) government in cross-border trade is more important
 - Balance of payments
 - System of accounts that records a country's international financial transactions. For example imported goods are done with own country carriers.²
 - Subsidies to marine merchant
 - Train their own merchant marine officers
 - Absorb some cost of building commercial vessels
 - Engage in activities to promote their own merchant fleets
 - Cargo preference rules for own country carriers

Macroenvironmental Influences on International Logistics (4 of 5)

- Economic factors that impact international logistics
 - **Currency fluctuations:** When one country's currency is weak relative to other currencies, it becomes more costly to import products, but exports often surge
 - **Market size:** population in China and India accounts for 1/3 of world population
 - **Income:** GDP per capita. Example low GDP per capita depicts single-use packets of products (small quantities)
 - **Infrastructure** for transportation including inland, air, and rail and for warehousing
 - **Economic integration** including free trade agreement and economic union (European union)

Macroenvironmental Influences on International Logistics (5 of 5)

- Cultural factors
 - Language
 - National holidays
 - Time orientation



Year	Beginning Date
2018	February 16
2019	February 5
2020	January 25
2021	February 12
2022	February 1
2023	January 22

International Documentation³ (1 of 2)

- Flow of documentation is as much a part of the main logistical flow as the flow of product
- Domestic shipments typically only require several pieces of documentation
- Export shipments typically require approximately 10 pieces of documentation
- Cross-border trades can require more than 100 separate documents
- Necessary documents are required at the point of importation (destination country)

³This material in this section is drawn from Donald F. Wood, Anthony Barone, Paul R. Murphy, and Daniel L. Wardlow, *International Logistics*, 2nd ed. (New York: Amacom, 2002), Chapter 12.

International Documentation⁴ (2 of 2)

Commonly used documents

- **Commercial invoices:** is similar to a domestic bill of lading summarizing transaction details and contains key information on good description, terms of sale and payment methods, shipment quantity, shipment method
- **Certificates of origin:** specifies country(ies) in which a product is manufactured and can be required by governments for control purposes or by an exporter
- **Shipper's export declaration (SED):** contains export transaction data such as the transportation mode(s), transaction participants, and description of content
- **Shipper's letter of instruction (SLI):** provides explicit shipment instructions indicating which parties should receive which documents, route of shipment and mode, what types of insurance and which insurance company(ies) to use

Terms of Sale (1 of 5)

- Terms of sale involves:
 - Parties working within the negotiations channel
 - Looking at the possible logistics channels
 - Determining when and where to transfer the following between buyer and seller:
 - Physical goods (logistics)
 - Payment for the goods, freight charges, and insurance for the in-transit goods (financing)
 - Legal title to the goods (ownership)
 - Required documentation
 - Responsibility for controlling or caring for the goods in transit (e.g., livestock)

Terms of Sale (2 of 5)

- Terms of sale for international shipments are commonly referred to as Incoterms
 - Developed in the 1930s by *the International Chamber of Commerce* and is periodically revised
 - Most recent revision, Incoterms 2010, reflects rapid expansion of global trade with focus on improved cargo security and new trends in cross-border transportation⁵

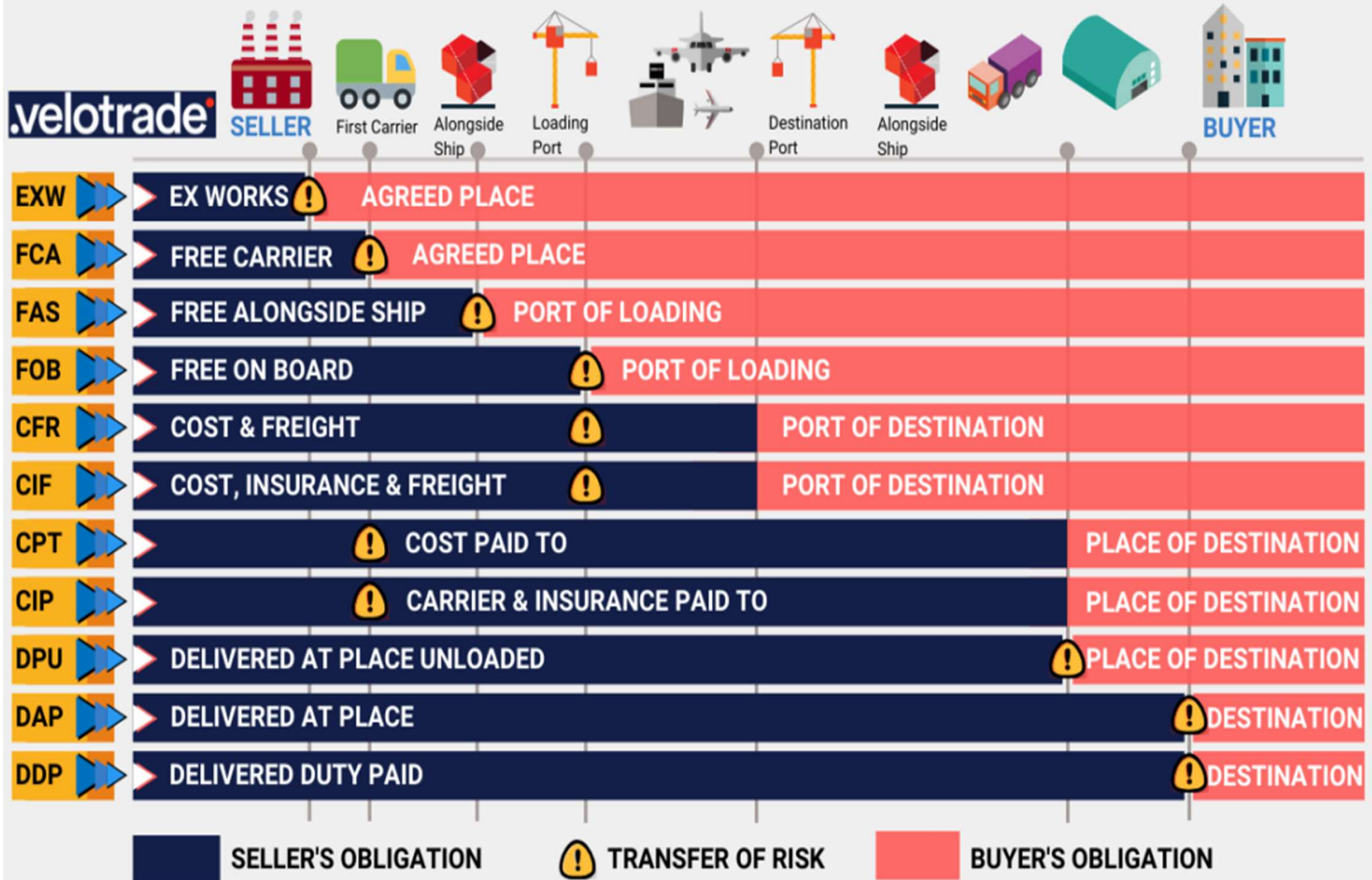
⁵Export.gov—Incoterms 2010.

Terms of Sale (3 of 5)

- Incoterms 2010 involve two key changes:
 - Organizing the terms by modes of transport
 - Characterized as Group 1
 - Applies to any mode of transport
 - Using the terms in both international and domestic transportation
 - Characterized as Group 2
 - Applies to sea and inland waterway transport only

INCOTERMS 2020

Point of Delivery and Transfer of Risk



Terms of Sale (4 of 5)

Group 1 Incoterms 2010

- **EXW (ExWorks):** seller fulfills his obligation by having the goods available for the buyer to pick up at his premises or another named place. Buyer bears all risk and costs starting when he picks up
- **FCA (Free Carrier):** Seller delivers the goods export cleared to the carrier chosen by the buyer. Buyer assumes all risks and costs with delivery of goods to final destination including transportation and customs fees
- **CPT (Carriage Paid To):** Similar to FCA but seller is not responsible for purchasing insurance
- **CIP (Carriage and Insurance Paid To):** Similar to FCA but seller purchases minimum insurance coverage

Terms of Sale (4 of 5)

Group 1 Incoterms 2010

- **DAT (Delivered at Terminal):** Seller clears the goods for export and bears all risks and costs associated with delivering the goods and unloading them at the terminal at named destination. Buyer is responsible for all costs and risks from this point forward including clearing the goods for import at the destination
- **DAP (Delivered at Place):** similar to DAT except the buyer is responsible for unloading the goods
- **DDP (Delivered Duty Paid):** The seller bears all risks and costs associated with delivering the goods to the named place of destination ready for unloading and clearing for import.

Terms of Sale (5 of 5)

Incoterms Group 2 terms

- **FAS (Free Alongside Ship):** Seller clears goods for export and delivers them ***alongside*** the vessel at shipment port. Buyer assumes all risks and costs from this point forward
- **FOB (Free on Board):** Seller clears the goods for export and delivers them ***onboard*** the vessel at shipment port. Buyer assumes all risks and costs from this point forward.
- **CFR (Cost and Freight):** Seller clears the goods for export and delivers them ***onboard*** the vessel at shipment port of shipment. The seller bears freight cost to destination port. Buyer assumes all risks for goods from delivered onboard

Terms of Sale (5 of 5)

Incoterms Group 2 terms

- **CIF (Cost, Insurance, and Freight):** Seller clears the goods for export and delivers them onboard the vessel at shipment port. Seller bears the cost of freight and insurance to the destination port. Seller's insurance requirement is only for **minimum coverage**. Buyer is responsible for all costs associated with unloading the goods at destination port and clearing goods for import. The risk passes from seller to buyer once the goods are onboard the vessel. at shipment port

Methods of Payment⁶ (1 of 2)

- Methods of payment refers to the manner by which a seller will be paid by a buyer
- Much more challenging and risky in international logistics than in domestic logistics
- Four methods of payment include:
 - Cash in advance: minimum risk to seller but major risk to buyer
 - Letters of credit: document issued by bank guaranteeing payment to a seller provided that seller has complied with the applicable terms and conditions of the particular transaction
 - Bills of exchange: binds buyer to pay a fixed sum of money to another party on demand or at a predetermined date involving a third party caller drawer
 - Open account: seller sends goods and documents to buyer who is entrusted to pay by a given date. This has maximum risk to seller but minimum risk to buyer

Figure 14.3 Letter of Credit

Source: DC.gov

DISTRICT OF COLUMBIA CAPTIVE INSURANCE COMPANY IRREVOCABLE LETTER OF CREDIT

Letter of Credit No. (00001)

A.B.C. Bank
Date Address
City, State

Commissioner
Department of Insurance, Securities and Banking
Attention: Joyce Lewis
810 First Street, NE, Suite 701
Washington, DC 20002

Commissioner:

1. We hereby establish our IRREVOCABLE LETTER OF CREDIT in your favor for the account of _____ up to the aggregate amount of _____ available by your draft(s) drawn on us, at sight, bearing the number of this IRREVOCABLE LETTER OF CREDIT No. (00001). This LETTER OF CREDIT shall expire at our Letter of Credit Department, _____ at our close of business on _____ unless as hereinafter extended.
2. This LETTER OF CREDIT is issued pursuant to the provisions of Section 7 (d) of the Captive Insurance Company Act of 2004, and on behalf of the above mentioned _____ (name of captive) which is applying for a certificate of authority to engage in the insurance business in the District of Columbia as a captive insurance company. We understand and agree that _____ (name of captive) has no obligation to reimburse us and we have no right of set off against any funds held by us for _____ (name of captive) in the event this LETTER OF CREDIT is drawn down, in whole or in part. By issuing this LETTER OF CREDIT, we waive any common law, statutory or contractual right of reimbursement or set off against _____ (name of captive) that may arise in the event this LETTER OF CREDIT is drawn down, in whole or in part.
3. It is a condition of the LETTER OF CREDIT that it shall be automatically extended for additional periods, each of one year, unless at least ninety calendar days prior to the then relevant expiration date we have advised you in writing that we elect not to extend. In that event, you may draw hereunder on or prior to the then relevant expiration date, up to the full amount then available hereunder, against your sight draft(s) on us, bearing the number of this LETTER OF CREDIT.
4. It is a further condition of this LETTER OF CREDIT that each automatic extension shall be measured from the then relevant expiration date, even though such date is not a business day in the District of Columbia for this Bank. It is also a condition of this LETTER OF CREDIT that, for the purpose of drawing hereunder, if the then relevant expiration date is a nonbusiness day for our Bank, drawing may be made not later than our next immediately following business day.
5. This LETTER OF CREDIT sets forth in full the terms of our undertaking, and such undertaking shall not in any way be modified, amended or amplified by reference to any note, document, instrument, statute, regulation or agreement referred to herein or in which this LETTER OF CREDIT is referred to or to which this LETTER OF CREDIT relates and any such reference shall not be deemed to incorporate herein by reference any note, document, instrument, statute, regulation, or agreement.
6. Each sight draft so drawn and presented shall be promptly honored by us if presented on or prior to the above stated expiration date or any extension thereof as above provided. Presentation under this LETTER OF CREDIT must be made at _____ located at _____ during normal banking hours.

Optional Provision

7. Unless otherwise expressly stated, this credit is subject to the Uniform Customs and Practices for Commercial Documentary credits fixed by the International Chamber of Commerce, ICC Publications No. 500 (1993 Rev.).

Very truly yours,

Methods of Payment⁷ (2 of 2)

- Payment method
 - Should be established at the time that a shipment price is decided upon
 - Can be influenced by key factors such as:
 - The country the product is to be sold in
 - The seller's assessment of buyer risk

⁷The material in this section is largely drawn from Jim Sherlock and Jonathan Reuvid, editors, *The Handbook of International Trade* (Edinburgh, UK: GMB Publishing, 2005), Part 8.

International Trade Specialists (1 of 10)

- International freight forwarders
 - Specialize in handling either vessel shipments or air shipments

International Trade Specialists (2 of 10)

- Principle functions of international freight forwarders include:
 - Advising on acceptance of letters of credit
 - Booking space on carriers
 - Preparing an export declaration
 - Preparing an air waybill or bill of lading
 - Obtaining consular documents
 - Arranging for insurance
 - Preparing and sending shipping notices and documents
 - Serving as general consultant on export matters

International Trade Specialists (3 of 10)

- Nonvessel-operating common carrier (NVOCC)
 - Often confused with the international freight forwarder
 - Consolidates freight from different shippers and leverages volume to negotiate favorable rates
- Export management company (EMC)
 - Firm that acts as the export sales department for a manufacturer
- Export packers
 - Custom pack shipments when the exporter lacks the equipment or the expertise to do so itself

International Trade Specialists (4 of 10)

- Transportation and inventory considerations in international logistics
 - Ocean shipping
 - Shipping conferences and alliances
 - International airfreight
 - Surface transport considerations

International Trade Specialists (5 of 10)

- Transportation and inventory considerations in international logistics
 - Ocean shipping
 - Approximately 60 percent of cross-border shipments move by water transportation
 - Variety of ship types include:
 - Dry-bulk cargo (grain, coal, sugar)
 - Specialized cargo
 - Liquid bulk cargo (oil transport)
 - Parcel tanker contains about 50 tanks of capacity 200 to 3,500 cubic feet
 - Containerships
 - Shipping conferences and alliances pool resources and extend market coverage

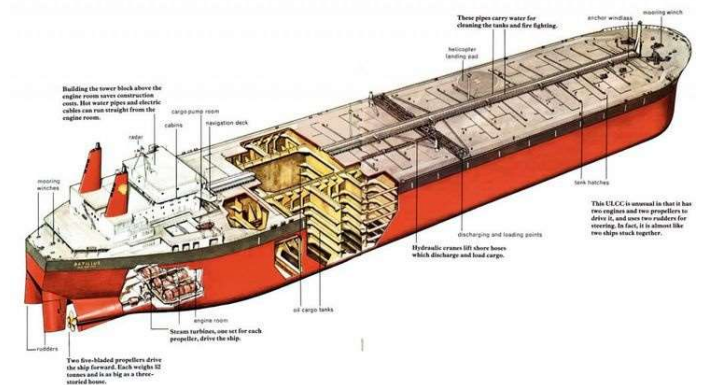


Figure 14.4: An Ocean Bulk Carrier Being Loaded with Export Coal Carried by a Mechanical Device at Far Left



Source: David Gaylor /Shutterstock



Oil tanker



RORO - Car Carriers



Chemical and gas tankers



Container vessels

Table 14.2: World's Busiest Container Ports (2014)

Port	Country	TEU Throughput
Shanghai	China	36,540,000
Singapore	Singapore	30,920,000
Shenzen	China	24,200,000
Ningbo	China	20,630,000
Hong Kong	China	20,070,000
Busan	South Korea	19,450,000
Qingdao	China	17,470,000
Guangzhou	China	17,220,000
Jebel Ali Dubai	United Arab Emirates	15,600,000
Tianjin	China	14,110,000

Source: Marsha Salisbury, Journal of Commerce, August 22, 2016.

International Trade Specialists (6 of 10)

- Transportation and inventory considerations in international logistics
 - Shipping conferences and alliances
 - Shipping conferences
 - Cartels of all ocean vessel operators operating between certain trade areas such as Asia and Europe
 - Provide rate stability and guaranteed space availability

International Trade Specialists (7 of 10)

- Transportation and inventory considerations in international logistics
 - Shipping conferences and alliances
 - Ocean carrier alliances
 - Carriers retain their individual identities but cooperate in the area of operations
 - Benefits include:
 - Sharing of vessel space
 - Ability to offer shippers a broader service network

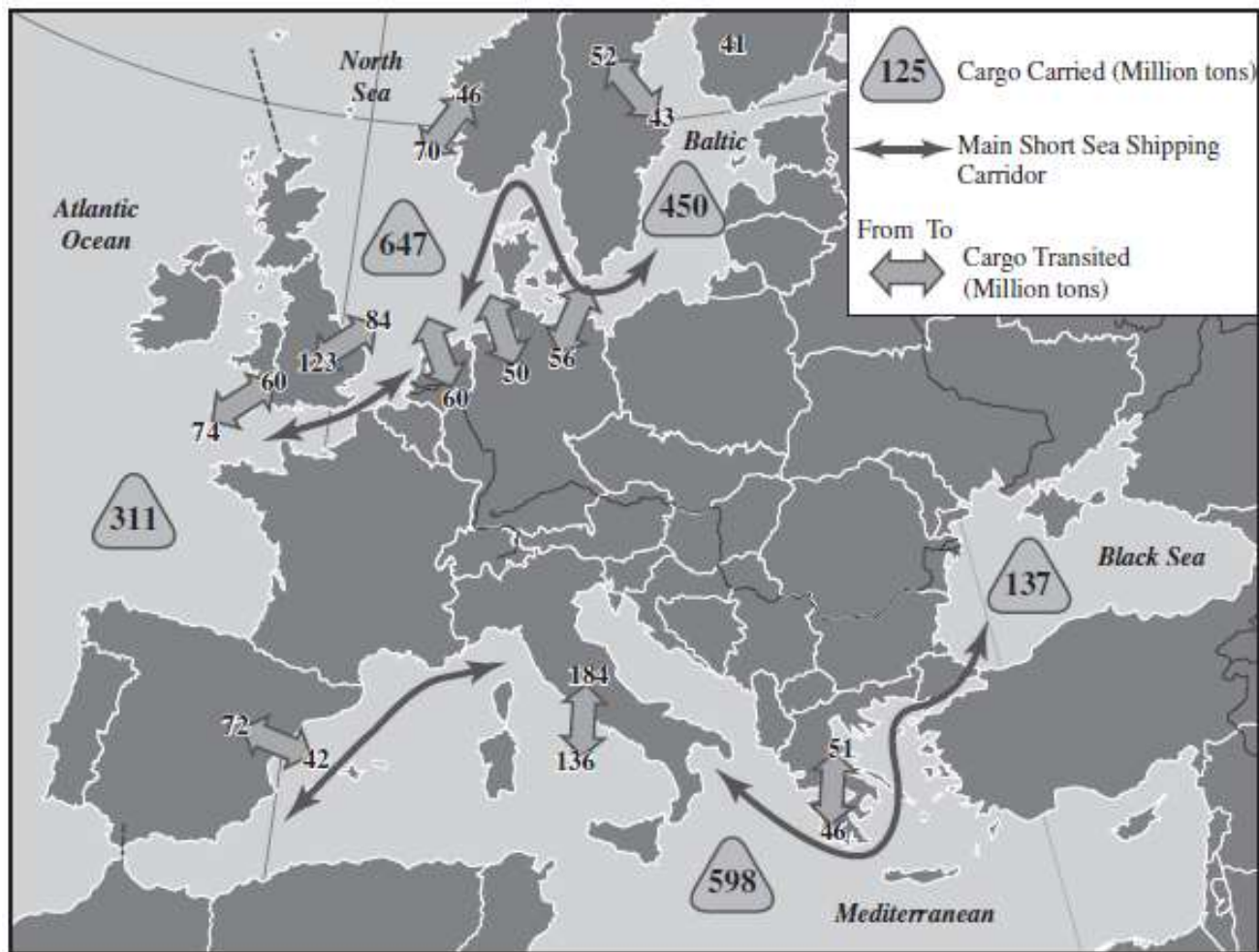
International Trade Specialists (8 of 10)

- Transportation and inventory considerations in international logistics
 - International airfreight
 - Three types of international airfreight operations include:
 - Chartered aircraft: renting an aircraft for back-to-back flights
 - Integrated air carriers: DHL, FedEx, UPS
 - Scheduled air carriers: in passengers airplane or in all-freight airplanes. Lufthanza flights non-stop freighters between Frankfurt and Shanghai

International Trade Specialists (9 of 10)

- Transportation and inventory considerations in international logistics
 - Surface transport considerations
 - Include an understanding of both a country's infrastructure and modal operating characteristics
 - Short-sea shipping (SSS), which refers to waterborne transportation that utilizes inland and coastal waterways to move shipments from domestic ports to their destination, is an alternative to surface transporting

Figure 14.5: The European Short Sea Shipping Network



Source: © Pearson Education.

International Trade Specialists (10 of 10)

- International trade inventories
 - Safety stocks must be large due to greater uncertainties, misunderstandings, and delays
 - Inventory valuation is difficult due to continually changing exchange rates
 - Product return (reverse logistics) policies must be understood
 - Warehousing facilities in developing countries may or may not meet the requirements of contemporary logistics systems

Logistics Performance Index (1 of 2)

- Emerging international logistics concept introduced in 2007
- Most recently updated in 2016
- Created in recognition of the importance of logistics in global trade
- Incorporates data for approximately 160 countries

Logistics Performance Index (2 of 2)

- Measures a country's performance across six logistical dimensions:
 - Efficiency of the clearance process by border control agencies, including customs
 - Quality of trade and transport related infrastructure
 - Ease of arranging competitively priced shipments
 - Competence and quality of logistics services
 - Ability to track and trace consignments
 - Timeliness of shipments in reaching destination within the scheduled or expected delivery time

LPI rankings for 2018

Score are out of 5

Country	Year	LPI Rank	LPI Score	Customs ?	Infrastructure ?	International shipments ?	Logistics competence ?	Tracking & tracing ?	Timeliness ?
Germany	2018	1	4.20	4.09	4.37	3.86	4.31	4.24	4.39
Sweden	2018	2	4.05	4.05	4.24	3.92	3.98	3.88	4.28
Belgium	2018	3	4.04	3.66	3.98	3.99	4.13	4.05	4.41
Austria	2018	4	4.03	3.71	4.18	3.88	4.08	4.09	4.25
Japan	2018	5	4.03	3.99	4.25	3.59	4.09	4.05	4.25
Netherlands	2018	6	4.02	3.92	4.21	3.68	4.09	4.02	4.25
Singapore	2018	7	4.00	3.89	4.06	3.58	4.10	4.08	4.32
Denmark	2018	8	3.99	3.92	3.96	3.53	4.01	4.18	4.41
United Kingdom	2018	9	3.99	3.77	4.03	3.67	4.05	4.11	4.33
Finland	2018	10	3.97	3.82	4.00	3.56	3.89	4.32	4.28
United Arab Emirates	2018	11	3.96	3.63	4.02	3.85	3.92	3.96	4.38
Saudi Arabia	2018	55	3.01	2.66	3.11	2.99	2.86	3.17	3.30

Country	LPI Rank	LPI Score	Customs ?	Infrastructure ?	International shipments ?	Logistics competence ?	Tracking & tracing ?	Timeliness ?
Germany	1	4.19	4.09	4.38	3.83	4.26	4.22	4.40
Netherlands	2	4.07	3.97	4.23	3.76	4.12	4.08	4.30
Sweden	3	4.07	3.95	4.22	3.88	4.04	4.02	4.32
Belgium	4	4.05	3.74	4.03	3.97	4.10	4.11	4.40
Singapore	5	4.05	4.00	4.14	3.72	4.08	4.05	4.34
United Kingdom	6	4.01	3.85	4.09	3.69	4.04	4.10	4.32
Japan	7	3.99	3.91	4.19	3.61	4.03	4.03	4.24
Austria	8	3.99	3.71	4.07	3.78	4.04	4.13	4.22
Hong Kong, China	9	3.96	3.85	4.02	3.85	3.94	3.95	4.18
United States	10	3.92	3.76	4.10	3.54	3.93	4.13	4.14
Denmark	11	3.92	3.88	3.89	3.59	3.98	3.94	4.26
Finland	12	3.92	3.89	3.95	3.56	3.88	4.10	4.17
Switzerland	13	3.91	3.75	4.07	3.57	3.92	4.02	4.20
United Arab Emirates	14	3.89	3.66	3.98	3.76	3.83	3.89	4.23
France	15	3.86	3.63	4.00	3.60	3.82	3.99	4.17
Luxembourg	16	3.84	3.67	3.84	3.68	3.83	3.78	4.27
Canada	17	3.81	3.70	3.91	3.45	3.90	3.91	4.03
Spain	18	3.78	3.57	3.79	3.72	3.78	3.78	4.04
Australia	19	3.77	3.76	3.92	3.40	3.76	3.83	4.00
Norway	20	3.74	3.62	3.84	3.48	3.75	3.83	3.96
Saudi Arabia	52	3.08	2.70	3.18	3.05	2.94	3.19	3.43

Aggregated LPI | Logistics Performance Index (worldbank.org)

Key Terms (1 of 3)

- Balance of payments
- Cargo preference
- Certificate of origin
- Commercial invoice
- Embargoes
- Export management company (EMC)
- Export packers
- Gross world product
- Import quota
- Incoterms
- International freight forwarders
- International logistics

Key Terms (2 of 3)

- Letter of credit
- Load centers
- Logistics Performance Index (LPI)
- Macroenvironmental influences
- Methods of payment
- Nontariff barriers
- Nonvessel-operating common carrier (NVOCC)
- Ocean carrier alliances
- Open account
- Open skies agreement

Key Terms (3 of 3)

- Shipper's export declaration (SED)
- Shipper's letter of instruction (SLI)
- Shipping conferences
- Short-sea shipping (SSS)
- Tariffs
- Terms of sale